IMPERIAL COLLEGE LONDON

Machine Learning in Finance

# Close-to-Open (C2O) Overnight Equity Strategy

Final Submission Report

| 250M NET RETURN | 250M NET VOL | 250M NET SHARPE | MAX DRAWDOWN |
|---|---|---|---|
| 7.31% | 4.97% | 1.445 | -10.19% |

**Decision:** final strategy is `phase2_g5_05_expanding`. The old 0.811 Sharpe strategy is retained only as previous-champion evidence.

Strategy universe: US Large-Cap Equities, 2010–2024 • Final strategy: volatility-scaled target with expanding-window weight estimation
AUM scenarios: $50M · $250M · $1B • Main headline AUM: $250M • Development cutoff: 31 December 2024

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## 1 Executive Summary and Coursework Map

The final strategy is `phase2_g5_05_expanding` : a daily, dollar-neutral, close-to-open long-short strategy using a volatility-scaled overnight-return target with expanding-window transparent feature-weight estimation. The target is:

```
target = overnight_next / (vol20 / sqrt(252))
```

where `overnight_next` is the close-to-next-open return and `vol20` is trailing 20-day close-to-close volatility, annualised and shifted by one trading day. The feature set is unchanged from the original point-in-time panel. The main research change is a risk-scaled label and an expanding training window, not a larger black-box model.

At the main headline AUM of 250M, the final strategy earns **7.31%** net annual return, **4.97%** net volatility, net Sharpe **1.445**, and max drawdown **-10.19%** over 2010–2024 after commission, auction slippage, borrow costs, and a 5% ADV20 participation cap.

We keep weak evidence in the report. The 2024 year is negative. The 2023–2024 holdout Sharpe is lower than validation. We do not have external prime-broker borrow validation. 1B capacity is clearly constrained. 2025–2026 remains held out for marker evaluation.

### Coursework Question Map

The brief contains **24 explicit report questions** across Sections 2–6:

| Brief Section | Topic | Explicit Questions | Where Answered Here |
|---|---|---|---|
| Section 2 | Daily panel questions | 6 | Data sources, return identity, earnings timing, short-interest lag, universe and stylised fact |
| Section 3 | Capacity-aware universe | 4 | Thresholds, participation cap, AUM capacity, binding constraints |
| Section 4 | Borrow filtering | 4 | Borrow proxy, external validation, tiered cost treatment, gross-to-net borrow impact |
| Section 5 | Alpha model | 5 | Information set, target, model/training, IC, ablation and weak periods |
| Section 6 | Portfolio and costs | 5 | Basket/weighting/turnover, 50M/250M/1B performance, cost drag, QuantStats, stress windows |

Section 7 adds sceptical-marker audit questions. This report answers those by tying look-ahead, capacity, borrow, robustness, and cost claims to frozen output files.

## 2 Methodology Pipeline

The pipeline is meant to be boring in the right places. It reads the same local data files, builds the same point-in-time panel, uses the same costs and capacity cap, and writes the same outputs every time. The only promoted modelling change is the volatility-scaled target with expanding-window estimation.

| Step | Stage | What the Code Does |
|---|---|---|
| 1 | Load and align data | Read prices, all_data, cheapness scores, earnings, short interest, GICS and benchmark files |
| 2 | Build point-in-time features | Lag close/high/low, accounting, valuation and short-interest fields; rank features cross-sectionally |
| 3 | Define investable universe | Use prior-year top 1000 market-cap names, then apply price, ADV, volatility, earnings and history filters |
| 4 | Train alpha by year | For each scored year, use only earlier years under the expanding-window protocol |
| 5 | Form daily baskets | Buy top 3% and short bottom 3%, with at least 15 names each side and equal weights before caps |
| 6 | Apply costs and capacity | Use commission, auction slippage, borrow tiers and 5% ADV20 per-name participation cap |
| 7 | Audit and report | Write performance, positions, IC, ablation, robustness, borrow, capacity and reproduction files |

The main implementation lives in `src/c2o_strategy/final_strategy.py` . The `Makefile` runs the reproduction in separate targets: final outputs, ablation, then report assets.

## 3 Data, Trading Clock, and Point-in-Time Panel

### 3.1 Data Source and Window

The pipeline uses the provided coursework data files under `data/`: daily adjusted prices, market cap, volume, GICS information, accounting/valuation variables, earnings calendar, short-interest proxies, regime data, and S&P 500 total return benchmark. The development sample runs from 1 January 2010 to 31 December 2024. No 2025 or 2026 observations are used.

The trading decision is made before the day-*t* close. Positions enter at the close and exit at the next open. Features using day-*t* close, high, low, accounting, or short-interest data are shifted unless they are known before the decision time. Cross-sectional ranks are formed within the same-day available cross-section only.

### 3.2 Return Reconciliation

The panel passes the open-close return identity check. We verify:

```
(1 + r_overnight) × (1 + r_intraday) − 1 = close_t / close_(t−1) − 1
```

on **5,469,968** stock-days at tolerance `1e-08`. The fail count is **0**, fail fraction is **0.000000**, and the maximum absolute residual is 4.44e-16.

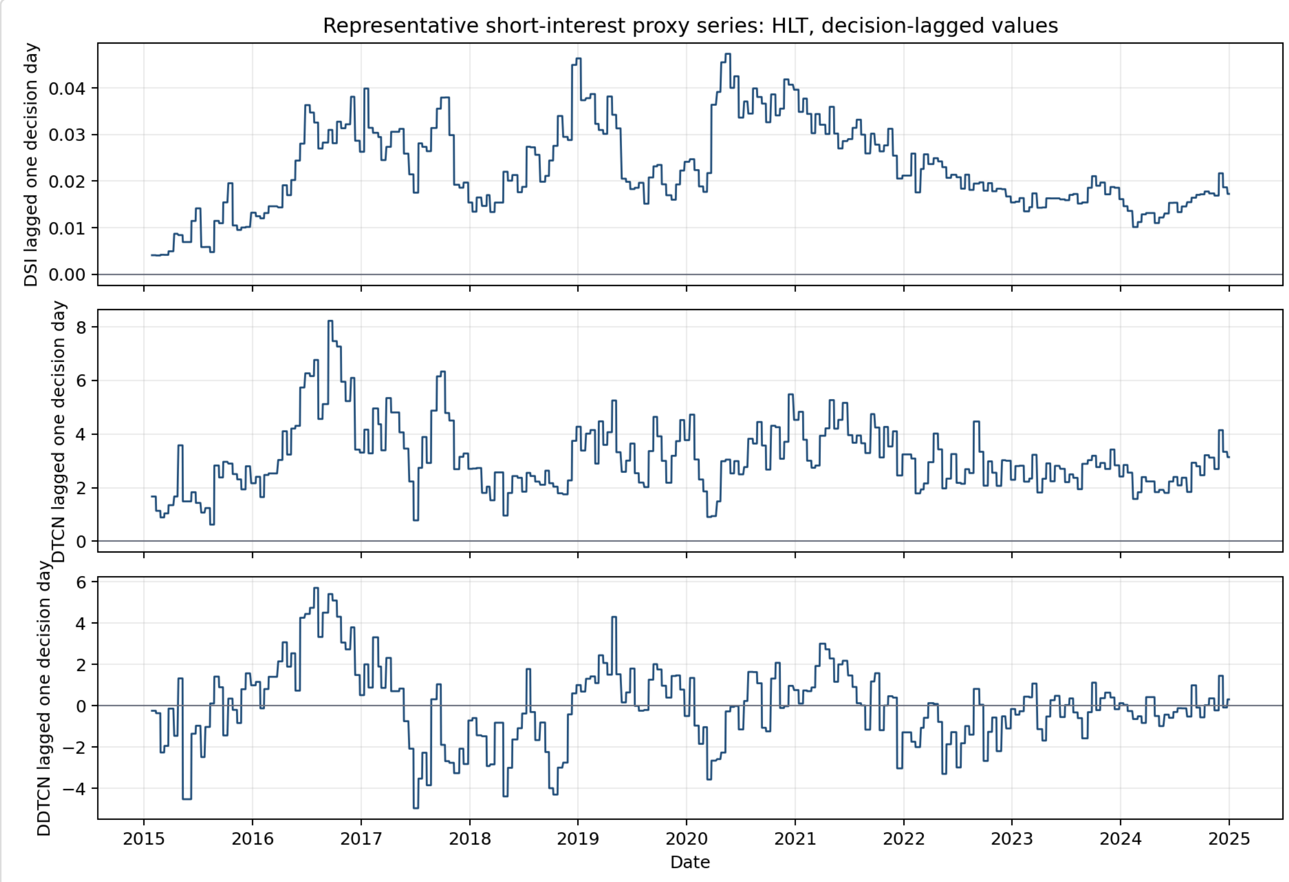
### 3.3 Earnings Timing and Short-Interest Lag

Earnings observations are mapped to a strategy trading date. A before-market announcement can be known before that day's close; an after-market announcement cannot be used for the same close decision. The pipeline excludes a ±1 trading-day window around the strategy date.

| Example | Ticker | Report Date | Timing | Strategy Date | Excluded Decision Dates |
|---|---|---|---|---|---|
| AMC example | MOS | 2010-01-05 | after | 2010-01-05 | 2010-01-04, 2010-01-05, 2010-01-06 |
| BMO example | RPM | 2010-01-06 | before | 2010-01-05 | 2010-01-04, 2010-01-05, 2010-01-06 |

Short-interest variables `dsi`, `dtcn`, and `ddtcn` are read from the provided point-in-time panel. We apply an additional one-trading-day decision lag before these fields enter alpha features or borrow-tier rules.

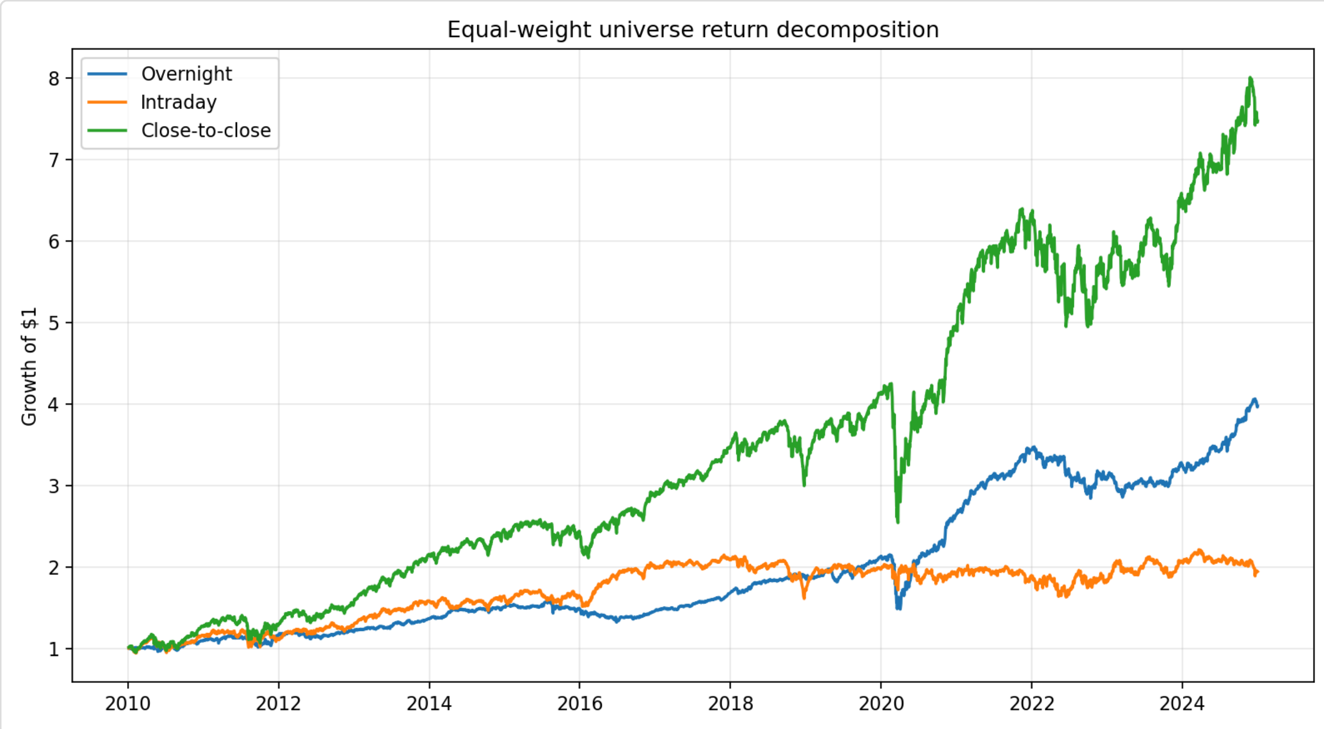
| Ticker | Source Feature Date | Decision Date | DSI | DTCN | DDTCN |
|---|---|---|---|---|---|
| HLT | 2015-01-27 | 2015-01-28 | 0.004 | 1.67 | -0.25 |
| HLT | 2015-01-28 | 2015-01-29 | 0.004 | 1.67 | -0.25 |
| HLT | 2015-01-29 | 2015-01-30 | 0.004 | 1.67 | -0.25 |




*Representative HLT short-interest proxy series, 2015–2024. Provided point-in-time proxies plus one-trading-day decision lag. No AUM or portfolio cost assumption.*

### 3.4 Stylised Fact Check

In our eligible equal-weight universe, the result is less clean than the textbook version: the return identity holds exactly, but close-to-close return is larger than overnight return because intraday returns are also positive. We use the overnight effect as motivation, not as proof that any strategy should work.




*Equal-weight eligible-universe overnight/intraday/close-to-close return decomposition, 2010–2024. No AUM, no strategy costs, annual large-cap universe.*

### 3.5 Universe Evolution

| Year | Universe Count | Reference Date | Median Year-Start Mcap | Mid-Year Exits |
|---|---|---|---|---|
| 2010 | 986 | 2009-12-31 | $2.46B | 0 |
| 2011 | 986 | 2010-12-31 | $3.25B | 0 |
| 2012 | 988 | 2011-12-30 | $3.25B | 0 |
| 2013 | 988 | 2012-12-31 | $3.86B | 0 |
| 2014 | 987 | 2013-12-31 | $5.42B | 0 |
| 2015 | 986 | 2014-12-31 | $6.07B | 0 |
| 2016 | 987 | 2015-12-31 | $6.06B | 0 |
| 2017 | 988 | 2016-12-30 | $7.07B | 0 |
| 2018 | 988 | 2017-12-29 | $8.73B | 0 |
| 2019 | 988 | 2018-12-31 | $7.47B | 0 |
| 2020 | 989 | 2019-12-31 | $9.62B | 0 |
| 2021 | 990 | 2020-12-31 | $11.62B | 0 |
| 2022 | 988 | 2021-12-31 | $14.16B | 0 |
| 2023 | 988 | 2022-12-30 | $11.50B | 0 |
| 2024 | 989 | 2023-12-29 | $13.06B | 0 |

## 4 Capacity-Aware Universe and Execution Assumptions

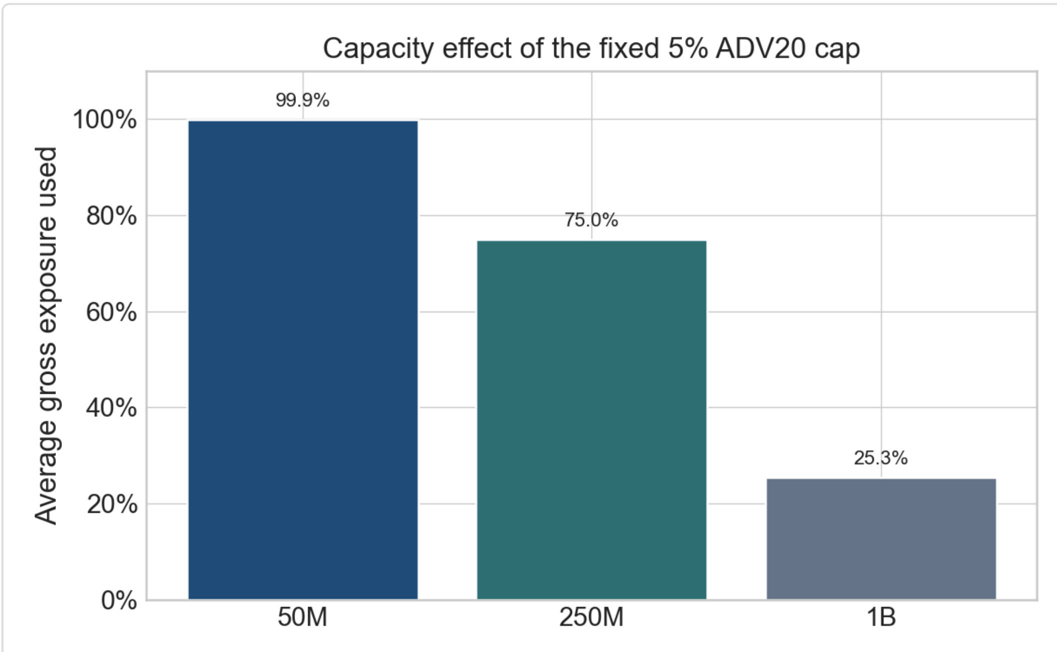
The investable universe uses simple filters: prior-year top 1000 market-cap membership, at least 252 history days, price above $5, ADV20 above $10M, annualised 20-day volatility between 5% and 120%, and a ±1 trading-day earnings exclusion window. The portfolio applies a 5% ADV20 per-name participation cap.

As a sanity check, a simple square-root proxy at the 5% ADV cap gives the following daily impact scale:

| Example | Ticker | Date | Market Cap | ADV20 | Vol20 | Daily Sigma | Participation | Impact bps |
|---|---|---|---|---|---|---|---|---|
| typical_mid_cap | EXE | 2024-12-31 | $23.07B | $220.6M | 18.69% | 1.18% | 5.00% | 18.4 |
| typical_large_cap | FISV | 2024-12-31 | $115.86B | $492.7M | 9.79% | 0.62% | 5.00% | 9.7 |

These proxy values are larger than the fixed 1.5 bps auction slippage per leg, so this is a conservative caveat.

| AUM | Avg Gross Exposure | Avg Abs Position | Max Abs Position | Avg Participation | Position-Days at Cap |
|---|---|---|---|---|---|
| 50M | 99.92% | $1.0M | $7.4M | 2.04% | 93.99% |
| 250M | 74.97% | $3.8M | $98.0M | 4.02% | 99.97% |
| 1B | 25.35% | $5.1M | $393.9M | 4.29% | 99.97% |




*Capacity by AUM for the final top/bottom 3% equal-weight strategy; 50M/250M/1B, fixed 5% ADV20 cap, full commission/slippage/borrow schedule.*

The 1B case is a capacity boundary case. Its average gross exposure is only **25.35%**, so the strategy cannot deploy target risk under the fixed 5% ADV20 cap.

### Binding Eligibility Reasons by Year

| Year | OK | ADV_FAIL | MCAP_FAIL | PRICE_FAIL | VOL_FAIL | EARN_WINDOW |
|---|---|---|---|---|---|---|
| 2010 | 154218 | 96974 | 46003 | 5995 | 307 | 2066 |
| 2011 | 165277 | 84995 | 52286 | 5303 | 355 | 3629 |
| 2012 | 165866 | 82506 | 59329 | 4103 | 661 | 3364 |
| 2013 | 184151 | 67903 | 69941 | 2766 | 392 | 3339 |
| 2014 | 204266 | 47834 | 82144 | 1369 | 915 | 4672 |
| 2015 | 211190 | 38791 | 93048 | 1350 | 326 | 7901 |
| 2016 | 218103 | 30514 | 101424 | 1115 | 905 | 8417 |
| 2017 | 223058 | 23282 | 110866 | 977 | 1372 | 8837 |
| 2018 | 234750 | 12302 | 119731 | 344 | 537 | 9592 |
| 2019 | 236744 | 10527 | 127586 | 915 | 468 | 9887 |
| 2020 | 226493 | 9139 | 134223 | 1386 | 12078 | 9976 |
| 2021 | 243163 | 2723 | 143615 | 48 | 811 | 10799 |
| 2022 | 242388 | 2717 | 148362 | 13 | 1332 | 11076 |
| 2023 | 241675 | 2026 | 151095 | 623 | 693 | 11465 |
| 2024 | 242803 | 1034 | 156668 | 0 | 1306 | 11905 |

## 5 Borrow Proxy and Short-Leg Financing

The final strategy uses tiered borrow costs, not a hard short exclusion. Tier A is 40 bps p.a. Tier B is 200 bps when `dsi_lag1 ≥ 0.08` , `dtcn_lag1 ≥ 5.0` , or `ddtcn_lag1 ≥ 1.0` . Tier C is 800 bps when `dsi_lag1 ≥ 0.15` , `dtcn_lag1 ≥ 10.0` , or both `dsi_lag1 ≥ 0.10` and `ddtcn_lag1 ≥ 1.5` .

At 250M, **56.79%** of selected short position-days are Tier B or C.

| Tier | Share of Short Days | Avg Short Notional | Total Borrow Cost |
|---|---|---|---|
| A | 43.21% | $4.4M | $2.8M |
| B | 38.25% | $3.6M | $10.2M |
| C | 18.54% | $2.8M | $15.4M |

### External Validation

We do not have direct prime-broker borrow-fee or securities-lending data. As partial external evidence, we checked our proxy against two well-documented short-squeeze episodes:

> **GME (2020-12-01 to 2021-02-15):** SEC staff report: GME short interest was around 100% of public float through most of 2020, reached 109.26% on 2020-12-31, and borrow fees were around 25% in January 2021 after exceeding 100% in Q2 2020.
> *Internal proxy:* Mean DSI = 0.938, Tier B/C fraction = 100.00%. Proxy flags the anecdotal crowded-short window as high borrow stress.

> **CVNA (2023-05-01 to 2023-07-31):** Reuters reported a May 2023 Carvana short-squeeze episode: the stock rose as much as 55%, short positions were about $488m, and Ortex described short sellers adding buy pressure while covering.
> *Internal proxy:* Mean DSI = 0.247, Tier B/C fraction = 100.00%. Proxy flags the anecdotal crowded-short window as high borrow stress.

### Hard-Exclusion Sensitivity

As a diagnostic, we remove selected shorts with `dsi_lag1 ≥ 10%` without replacement. This is not the promoted strategy.

| AUM | Final Sharpe | Hard-Excl Sharpe | Final Return | Final Max DD |
|---|---|---|---|---|
| 50M | 1.149 | 1.363 | 5.72% | -14.33% |
| 250M | 1.240 | 1.462 | 5.98% | -10.43% |
| 1B | 1.125 | 1.383 | 2.89% | -5.63% |

The result supports our choice to use tiered borrow costs rather than hard exclusion. High-DSI shorts are not the only source of performance.

## 6 Baseline, Promotion Decision, and Challenger Models

The first baseline predicted raw next overnight return. The promoted final model keeps the volatility-scaled target and changes the training window to expanding.

| Model | 250M Sharpe | 250M Return | Max Drawdown | Description |
|---|---|---|---|---|
| Previous champion | 0.811 | 3.93% | -10.19% | 4y rolling vol-scaled target |
| Final champion | 1.445 | 7.31% | -10.19% | expanding-window vol-scaled target |

The promotion rule was stricter than "pick the largest Sharpe". We wanted validation improvement, non-failure in 2023–2024, no obvious drawdown damage, and no hidden relaxation of costs or capacity.

### Top Phase-2 Challengers

| Experiment | Full Sharpe | Full Max DD | Passes Rule |
|---|---|---|---|
| phase2_g5_05_expanding | 1.445 | -10.19% | True |
| phase2_g2_06_score_div_vol_liquidity_floor | 1.276 | -8.80% | True |
| phase2_g2_04_score_divided_by_volatility | 1.241 | -8.74% | True |
| phase2_g6_06_elastic_net_ranked_features | 1.258 | -11.89% | True |
| phase2_g2_03_score_weighted | 1.088 | -8.75% | True |

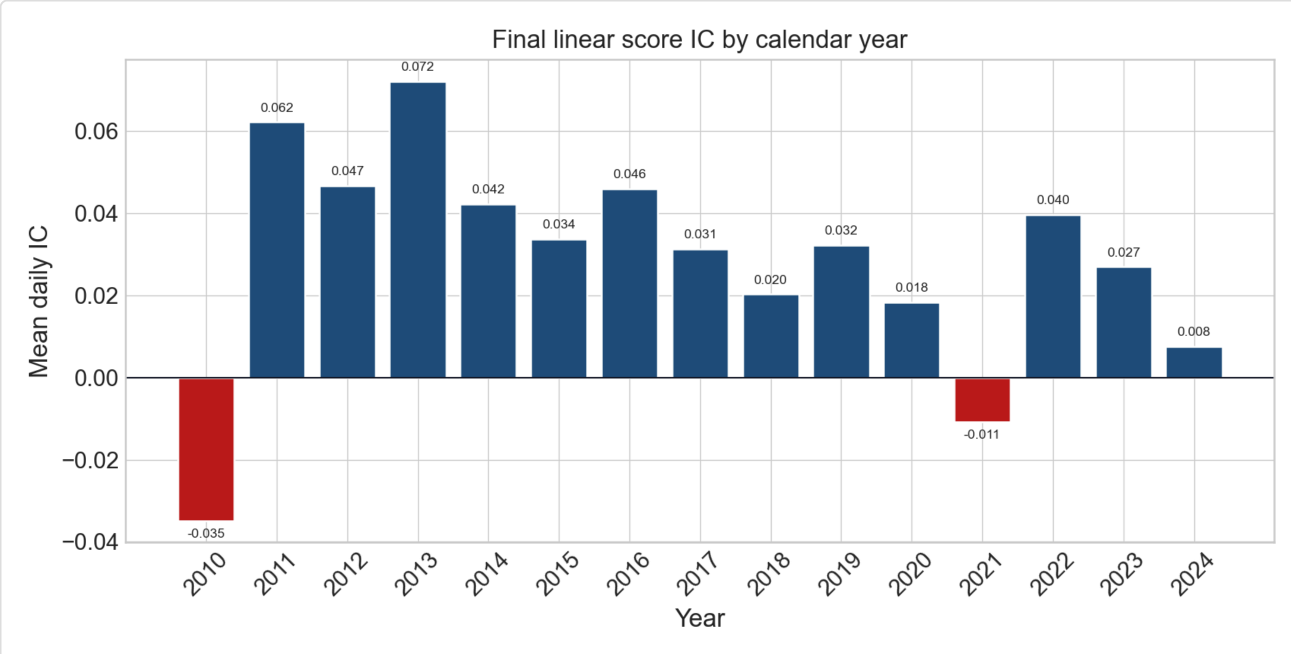
## 7 Alpha Design, Model Selection, and Interpretability

### 7.1 Target and Model Class

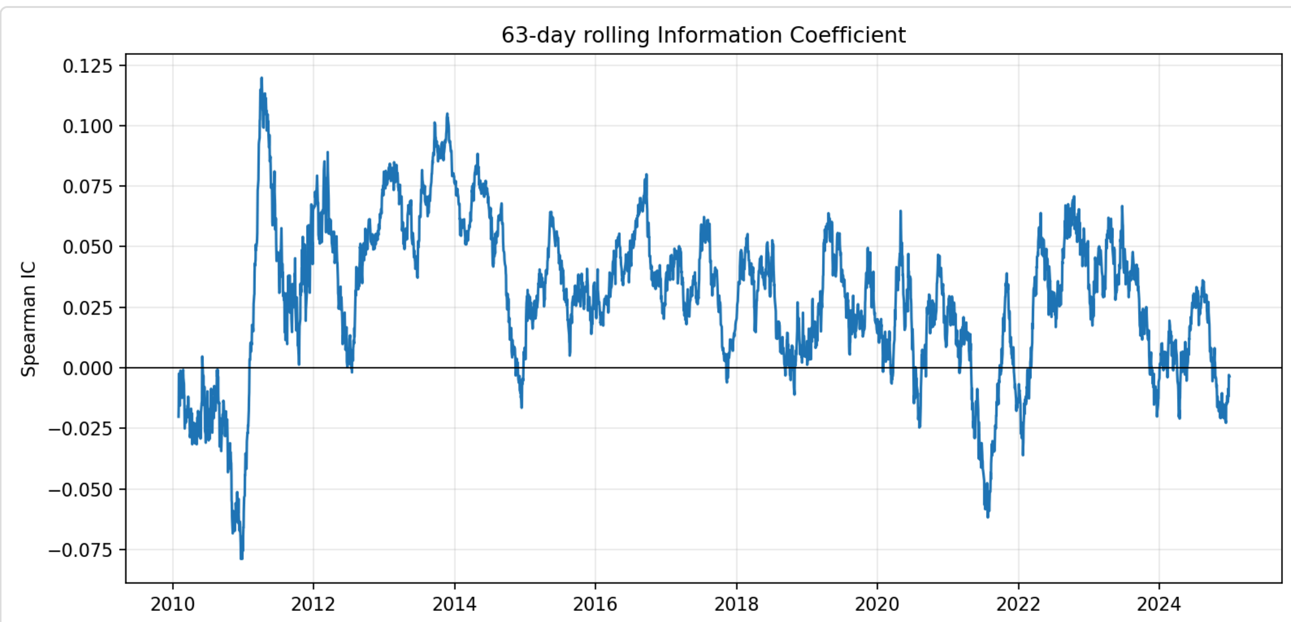
The baseline predicted raw next overnight return. That target is noisy because high-volatility names dominate. The final target divides by trailing daily volatility, so the model asks which stocks have better expected overnight return per unit of recent risk. The alpha learner is a 50% prior and 50% learned feature-weight correlation estimate on ranked point-in-time features.

### 7.2 Information Coefficient and Weak Regimes

The final alpha has full-sample mean IC **0.029** with t-stat **9.93**. Yearly IC is not uniformly positive. The weak years: 2010 has negative IC, 2022 has negative IC, and 2024 has near-zero IC.

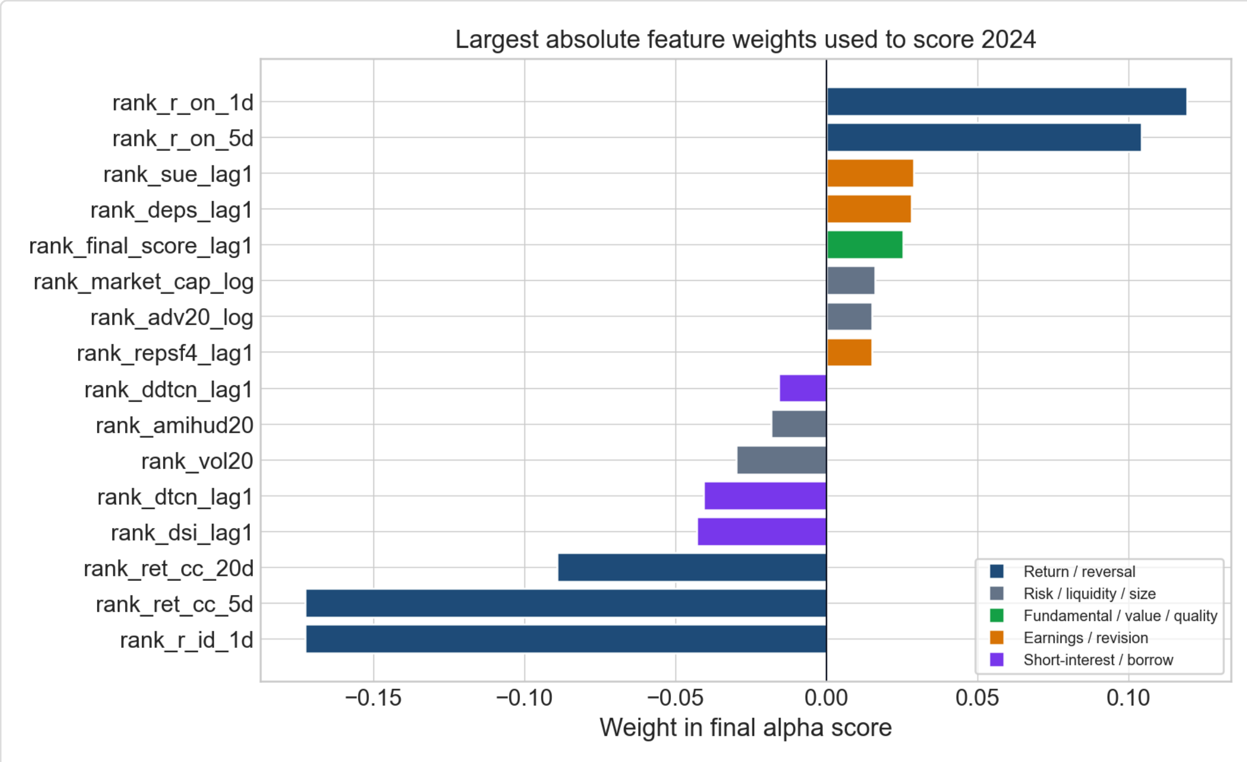



*Final linear-score mean IC by calendar year, 2010–2024. Negative years highlighted. 250M AUM, top/bottom 3% basket, 5% ADV20 cap, Section 6.3 cost schedule.*




*63-day rolling IC for the final linear score. Spearman correlation between alpha score and subsequent overnight return, 2010–2024.*

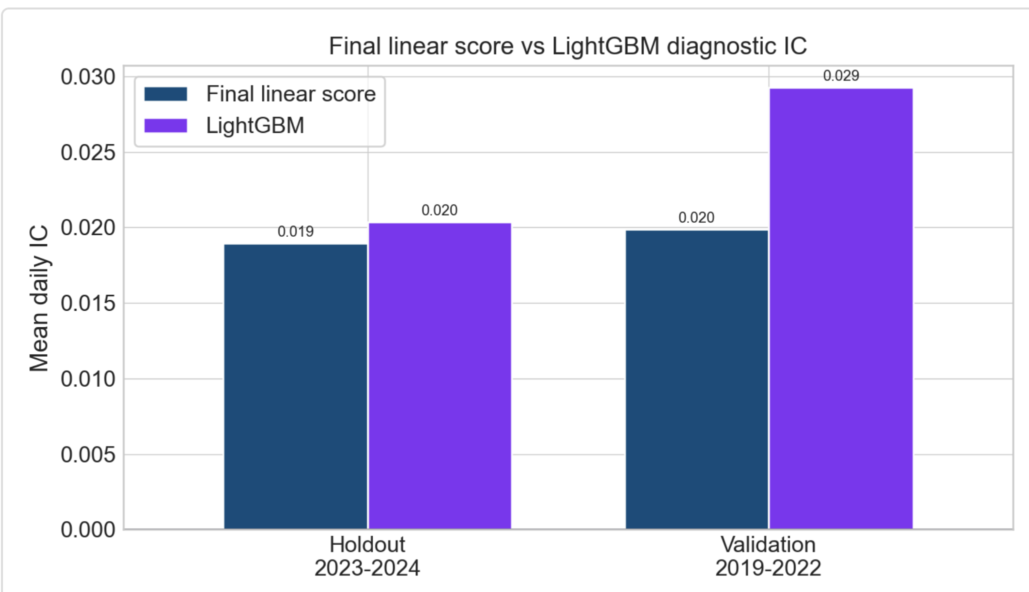
### 7.3 Feature Weights




*Largest 2024 final linear-score feature weights. Transparent expanding-window model, unchanged feature set, no LightGBM portfolio promotion.*

### 7.4 LightGBM Diagnostic

LightGBM was run as an optional IC diagnostic. It is not the final strategy.

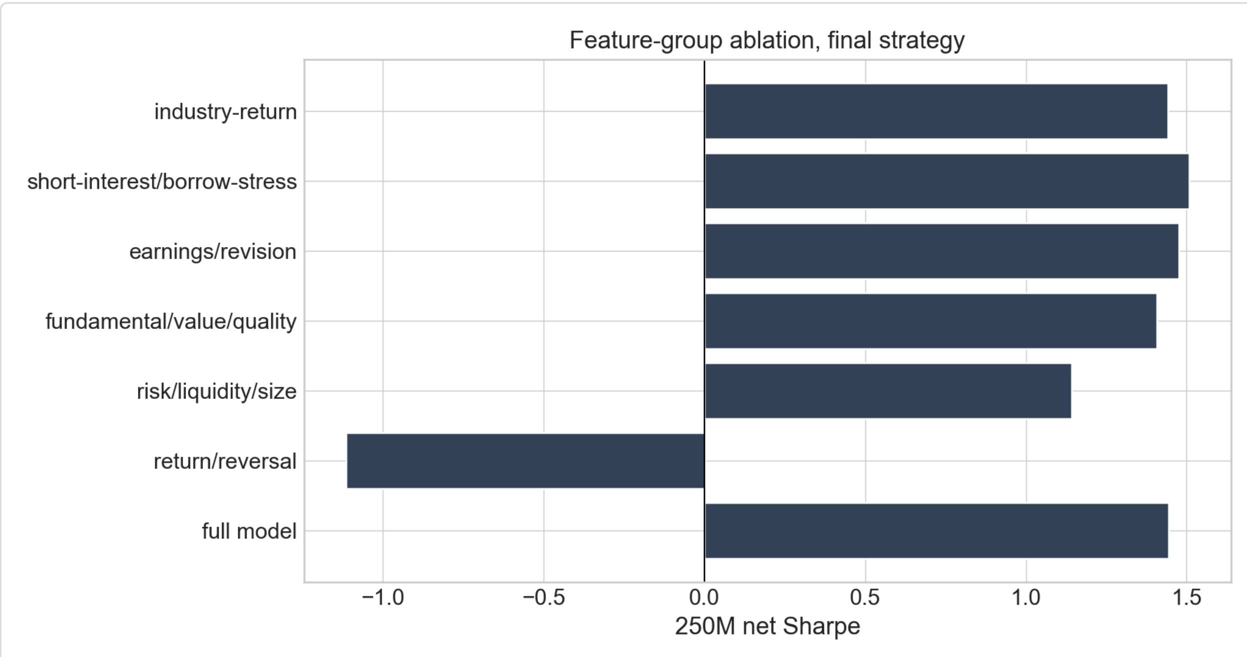



*Final linear score versus LightGBM diagnostic IC, validation and internal holdout. Same data panel and cutoff. LightGBM is not promoted as a costed strategy.*

### 7.5 Feature Ablation

Removing return/reversal destroys the strategy. Other groups are less clean: removing earnings/revision or short-interest/borrow-stress can improve frozen Sharpe. We interpret those variables partly as risk, cost, crowding, or regime controls.

| Variant | Removed Group | IC Mean | IC t-stat | Net Return | Net Vol | Net Sharpe | Max DD |
|---|---|---|---|---|---|---|---|
| full_model | none | 0.029 | 9.93 | 7.31% | 4.97% | 1.445 | -10.19% |
| remove_return/reversal | return/reversal | 0.011 | 3.33 | -3.62% | 3.26% | -1.114 | -46.66% |
| remove_risk/liquidity/size | risk/liquidity/size | 0.028 | 10.86 | 5.89% | 5.12% | 1.144 | -8.24% |
| remove_fundamental/value/quality | fundamental/value/quality | 0.030 | 10.20 | 7.01% | 4.90% | 1.407 | -9.75% |
| remove_earnings/revision | earnings/revision | 0.028 | 9.75 | 7.31% | 4.86% | 1.477 | -10.19% |
| remove_short-interest/borrow-stress | short-interest/borrow-stress | 0.029 | 9.85 | 7.70% | 5.00% | 1.508 | -11.32% |
| remove_industry-return | industry-return | 0.029 | 9.94 | 7.27% | 4.96% | 1.442 | -10.19% |




*Feature-group ablation at 250M for the final top/bottom 3% equal-weight strategy; fixed 5% ADV20 cap and full cost schedule.*

## 8 Portfolio Construction, Costs, and Headline Results

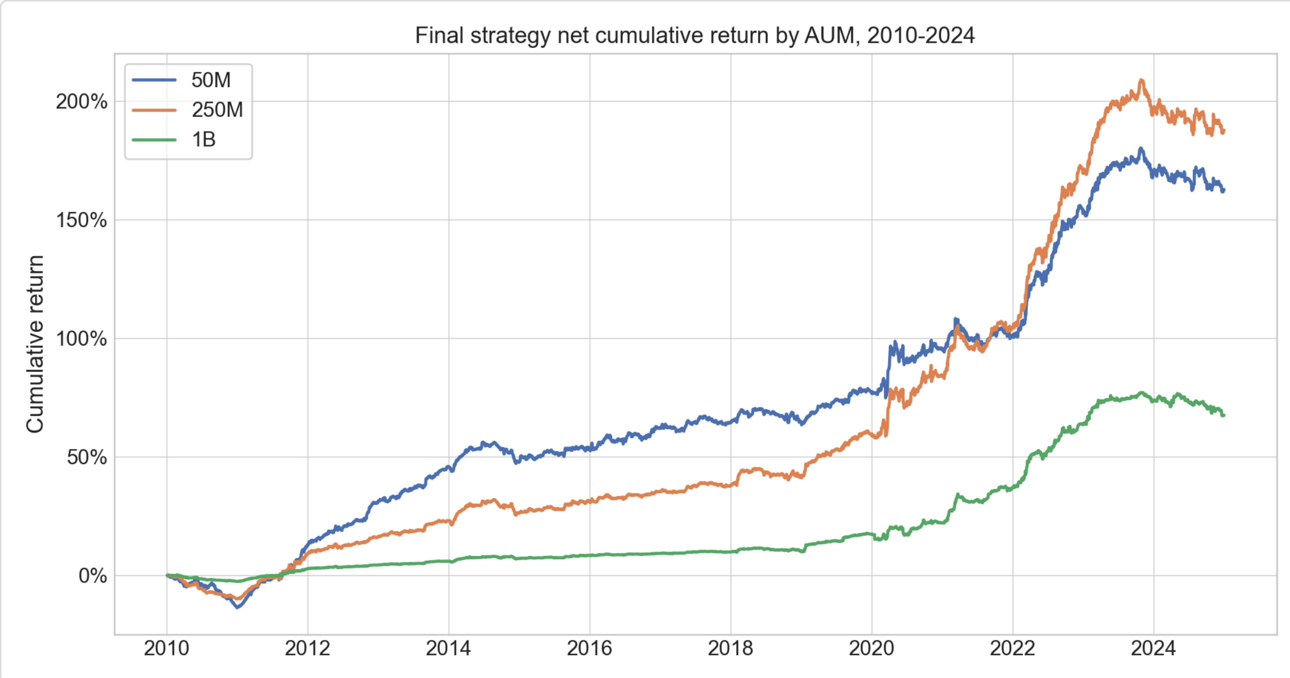
Each day the strategy ranks eligible stocks by raw alpha score, selects the top and bottom 3% with a minimum of 15 names per side, equal-weights each side, then applies per-name capacity caps. The resulting book is dollar-neutral after capacity sizing. Average selected names are about 24.9 long and 24.9 short per day.

| AUM | Net Return | Net Vol | Sharpe | Max DD | Turnover | Gross Exposure | Long/Short Names |
|---|---|---|---|---|---|---|---|
| 50M | 6.66% | 5.03% | 1.308 | -13.82% | 1.998 | 99.92% | 24.9/24.9 |
| 250M | 7.31% | 4.97% | 1.445 | -10.19% | 1.499 | 74.97% | 24.9/24.9 |
| 1B | 3.50% | 2.69% | 1.293 | -5.56% | 0.507 | 25.35% | 24.9/24.9 |

### Gross-to-Net Sharpe Decomposition

| AUM | Gross Sharpe | Commission Drag | Slippage Drag | Borrow Drag | Net Sharpe | Avg Borrow bps/day |
|---|---|---|---|---|---|---|
| 50M | 3.539 | 0.501 | 1.503 | 0.227 | 1.308 | 0.451 |
| 250M | 3.112 | 0.378 | 1.137 | 0.152 | 1.445 | 0.301 |
| 1B | 2.321 | 0.233 | 0.704 | 0.091 | 1.293 | 0.098 |

Auction slippage is the largest explicit cost drag. The QuantStats tear-sheet is provided as `outputs/quantstats_250m.html` against the S&P 500 total return benchmark.




*Final net cumulative returns by AUM; 50M/250M/1B, final top/bottom 3% equal-weight strategy, fixed 5% ADV20 cap and full commission/slippage/borrow schedule.*

## 9 Robustness and Stress Tests

### Stress Windows

| Window | Return | Vol | Sharpe | Max DD | Gross Exposure |
|---|---|---|---|---|---|
| late_2018 | -2.35% | 4.71% | -0.482 | -1.92% | 85.28% |
| 2020_q1 | 34.41% | 10.78% | 2.799 | -4.14% | 88.95% |
| 2022_drawdown | 32.24% | 8.02% | 3.525 | -2.63% | 99.23% |

### Pre/Post Split

| Window | Return | Vol | Sharpe | Max DD | Gross Exposure |
|---|---|---|---|---|---|
| 2010_2016 | 4.42% | 3.45% | 1.273 | -10.19% | 60.48% |
| 2017_2024 | 9.90% | 5.99% | 1.608 | -7.64% | 87.66% |

Tail diagnostics at 250M: worst 3-month return **-4.46%**, worst 6-month return **-7.31%**, worst 12-month return **-10.09%**. Top 5% best days contribute **1.461** times total PnL. Lag-1 return autocorrelation is **-0.007**.

### Lo Autocorrelation-Adjusted Sharpe

| AUM | Raw Sharpe | Lo-Adjusted Sharpe | Lag-1 Autocorrelation |
|---|---|---|---|
| 50M | 1.308 | 1.300 | -0.0137 |
| 250M | 1.445 | 1.423 | -0.0065 |
| 1B | 1.293 | 1.254 | -0.0107 |

## 10 Reproducibility and Code Audit

The full pipeline is deterministic. Random sampling is not used in the final alpha learner. The report-ready output files are generated from code, not edited by hand. The local reproduction commands are:

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt
PYTHON=.venv/bin/python make reproduce-final
PYTHON=.venv/bin/python make ablation
PYTHON=.venv/bin/python make report-assets
.venv/bin/python -m pytest -q
```

The tests cover portfolio neutrality, submission rules, capacity fields, and the blended challenger logic. The notebook `notebooks/c2o_research_process_and_tests.ipynb` reproduces the key report checks from generated outputs.

## 11 Limitations and Next Work

| Area | Limitation | How We Handle It |
|---|---|---|
| Borrow data | No external prime-broker feed available | Tiered borrow as proxy plus hard-exclusion sensitivity |
| Market impact | Backtest uses fixed auction slippage | Disclose square-root impact proxy as caveat |
| Holdout decay | 2023-2024 Sharpe lower than validation | Keep the number visible; avoid overclaiming |
| Capacity | 1B cannot deploy target gross exposure | Use 250M as headline; treat 1B as boundary |
| Model complexity | More aggressive ML models untested fully | Keep ridge/elastic net/blend as challengers |

The next modelling direction is the ensemble, not a sudden jump to a black-box model. The fixed blend already improves the 2023-2024 holdout Sharpe, but it does not satisfy the full promotion rule.

## 12 Sceptical Marker Checklist

**Look-ahead.** Features are observable by the decision time or shifted. Earnings use strategy trading dates with AMC/BMO treatment. Short interest is lagged an extra trading day. The universe is prior-year-end defined. No 2025+ observations enter development.

**Statistical robustness.** We report validation, internal holdout, year-by-year results, IC by year/regime, stress windows, rolling loss windows, lag-1 autocorrelation, and PnL concentration. We do not hide the lower 2023-2024 holdout Sharpe or negative 2024 return.

**Capacity and execution honesty.** The 5% ADV20 cap is binding frequently. At 1B, average gross exposure drops to 25.35%, so 1B is a capacity boundary, not the main headline.

**Borrow honesty.** Borrow costs are tiered and charged daily. No external borrow fee validation is claimed beyond anecdotal evidence (GME, CVNA). A DSI≥10% hard-exclusion diagnostic is reported and does not break the strategy.

**Reporting integrity.** All headline tables and charts trace to code-generated outputs. The report uses 250M as the main case but also discloses 50M and 1B. Figures are labelled with the cost and capacity context.

## 13 Conclusion

The final submitted strategy is the expanding-window volatility-scaled close-to-open alpha. It is more aggressive than the plain baseline because the target is risk-scaled and the challenger search considered ridge, elastic net, and blended ML scores. It is still controlled enough for submission because the promoted model is auditable, deterministic, point-in-time, and checked under costs, borrow, capacity, ablation, and stress tests. The next research direction is an ensemble challenger, but only if it improves validation and 2023-2024 holdout without worsening drawdown.

## A Appendix A: Universe and Eligibility Evidence

| Year | Universe Count | Reference Date | Median Year-Start Mcap | Mid-Year Exits |
|---|---|---|---|---|
| 2010 | 986 | 2009-12-31 | $2.46B | 0 |
| 2011 | 986 | 2010-12-31 | $3.25B | 0 |
| 2012 | 988 | 2011-12-30 | $3.25B | 0 |
| 2013 | 988 | 2012-12-31 | $3.86B | 0 |
| 2014 | 987 | 2013-12-31 | $5.42B | 0 |
| 2015 | 986 | 2014-12-31 | $6.07B | 0 |
| 2016 | 987 | 2015-12-31 | $6.06B | 0 |
| 2017 | 988 | 2016-12-30 | $7.07B | 0 |
| 2018 | 988 | 2017-12-29 | $8.73B | 0 |
| 2019 | 988 | 2018-12-31 | $7.47B | 0 |
| 2020 | 989 | 2019-12-31 | $9.62B | 0 |
| 2021 | 990 | 2020-12-31 | $11.62B | 0 |
| 2022 | 988 | 2021-12-31 | $14.16B | 0 |
| 2023 | 988 | 2022-12-30 | $11.50B | 0 |
| 2024 | 989 | 2023-12-29 | $13.06B | 0 |

| Year | OK | ADV_FAIL | MCAP_FAIL | PRICE_FAIL | VOL_FAIL | EARN_WINDOW |
|---|---|---|---|---|---|---|
| 2010 | 154218 | 96974 | 46003 | 5995 | 307 | 2066 |
| 2011 | 165277 | 84995 | 52286 | 5303 | 355 | 3629 |
| 2012 | 165866 | 82506 | 59329 | 4103 | 661 | 3364 |
| 2013 | 184151 | 67903 | 69941 | 2766 | 392 | 3339 |
| 2014 | 204266 | 47834 | 82144 | 1369 | 915 | 4672 |
| 2015 | 211190 | 38791 | 93048 | 1350 | 326 | 7901 |
| 2016 | 218103 | 30514 | 101424 | 1115 | 905 | 8417 |
| 2017 | 223058 | 23282 | 110866 | 977 | 1372 | 8837 |
| 2018 | 234750 | 12302 | 119731 | 344 | 537 | 9592 |
| 2019 | 236744 | 10527 | 127586 | 915 | 468 | 9887 |
| 2020 | 226493 | 9139 | 134223 | 1386 | 12078 | 9976 |
| 2021 | 243163 | 2723 | 143615 | 48 | 811 | 10799 |
| 2022 | 242388 | 2717 | 148362 | 13 | 1332 | 11076 |
| 2023 | 241675 | 2026 | 151095 | 623 | 693 | 11465 |
| 2024 | 242803 | 1034 | 156668 | 0 | 1306 | 11905 |

## B Appendix B: Feature Inventory and IC by Year

| Feature | Formula | Lag | Observable 15:50 ET |
|---|---|---|---|
| rank_r_on_1d | r_overnight = adj_open_t / adj_close_{t-1} - 1 | 0 trading days; Open_t is known after the 09:30 auction | Yes |
| rank_r_on_5d | rolling mean of r_overnight over 5 sessions, min_periods=3 | 0 trading days; uses overnight returns available by 15:50 ET | Yes |
| rank_r_id_1d | r_intraday_{t-1} = adj_close_{t-1} / adj_open_{t-1} - 1 | 1 trading day | Yes |
| rank_ret_cc_5d | adj_close_{t-1} / adj_close_{t-6} - 1 | 1 trading day | Yes |
| rank_ret_cc_20d | adj_close_{t-1} / adj_close_{t-21} - 1 | 1 trading day | Yes |
| rank_ret_cc_60d | adj_close_{t-1} / adj_close_{t-61} - 1 | 1 trading day | Yes |
| rank_vol20 | annualised rolling std of close-to-close returns over 20 sessions, shifted 1 day | 1 trading day | Yes |
| rank_vol60 | annualised rolling std of close-to-close returns over 60 sessions, shifted 1 day | 1 trading day | Yes |
| rank_amihud20 | rolling mean over 20 sessions of abs(r_close_close) / dollar_volume, shifted 1 day | 1 trading day | Yes |
| rank_adv20_log | log(1 + trailing 20-day average dollar volume), shifted 1 day | 1 trading day | Yes |
| rank_market_cap_log | log(1 + market_cap_{t-1}) | 1 trading day | Yes |
| rank_piot_norm_lag1 | piot_norm_{t-1} | 1 trading day | Yes |
| rank_gross_profit_margin_lag1 | gross_profit_margin_{t-1} | 1 trading day | Yes |
| rank_asset_turnover_ratio_lag1 | asset_turnover_ratio_{t-1} | 1 trading day | Yes |
| rank_net_debt_to_equity_lag1 | net_debt_to_equity_{t-1} | 1 trading day | Yes |
| rank_price_to_book_lag1 | price_to_book_{t-1} | 1 trading day | Yes |
| rank_ev_to_ebit_lag1 | ev_to_ebit_{t-1} | 1 trading day | Yes |
| rank_sue_lag1 | sue_{t-1} | 1 trading day | Yes |
| rank_deps_lag1 | deps_{t-1} | 1 trading day | Yes |
| rank_reps1_lag1 | reps1_{t-1} | 1 trading day | Yes |
| rank_repsf4_lag1 | repsf4_{t-1} | 1 trading day | Yes |
| rank_final_score_lag1 | final_score_{t-1} | 1 trading day | Yes |
| rank_score_velocity_lag1 | score_velocity_{t-1} | 1 trading day | Yes |
| rank_momentum_score_lag1 | momentum_score_{t-1} | 1 trading day | Yes |
| rank_dsi_lag1 | dsi_{t-1}; daily short-interest proxy after publication/vendor lag already in source | 1 trading day after source availability | Yes, if source obeys Section 2.1.3 lag convention |
| rank_dtcn_lag1 | dtcn_{t-1}; daily days-to-cover proxy after publication/vendor lag already in source | 1 trading day after source availability | Yes, if source obeys Section 2.1.3 lag convention |
| rank_ddtcn_lag1 | ddtcn_{t-1}; lagged change/stress proxy after publication/vendor lag already in source | 1 trading day after source availability | Yes, if source obeys Section 2.1.3 lag convention |
| rank_industry_return_lag1 | industry_return_{t-1} | 1 trading day | Yes |

| Year | Mean IC | IC t-stat | Days | Median N |
|---|---|---|---|---|
| 2010.0 | -0.0348 | -2.61 | 252.0 | 608.5 |
| 2011.0 | 0.0622 | 3.70 | 252.0 | 663.0 |
| 2012.0 | 0.0467 | 4.09 | 250.0 | 672.0 |
| 2013.0 | 0.0722 | 9.68 | 252.0 | 738.0 |
| 2014.0 | 0.0423 | 4.71 | 252.0 | 816.0 |
| 2015.0 | 0.0338 | 3.93 | 252.0 | 853.5 |
| 2016.0 | 0.0460 | 4.94 | 252.0 | 878.0 |
| 2017.0 | 0.0313 | 3.75 | 251.0 | 911.0 |
| 2018.0 | 0.0203 | 1.86 | 251.0 | 958.0 |
| 2019.0 | 0.0323 | 2.97 | 252.0 | 962.0 |
| 2020.0 | 0.0183 | 1.45 | 253.0 | 947.0 |
| 2021.0 | -0.0107 | -0.94 | 252.0 | 993.0 |
| 2022.0 | 0.0397 | 3.20 | 251.0 | 994.0 |
| 2023.0 | 0.0270 | 2.32 | 250.0 | 998.0 |
| 2024.0 | 0.0076 | 0.77 | 251.0 | 997.0 |

## C Appendix C: 250M Year-by-Year Performance

| Year | Return | Vol | Sharpe | Max DD | Gross Exposure |
|---|---|---|---|---|---|
| 2010 | -10.09% | 3.42% | -3.094 | -10.19% | 42.46% |
| 2011 | 21.39% | 5.63% | 3.474 | -2.08% | 67.54% |
| 2012 | 6.70% | 2.80% | 2.333 | -1.68% | 56.93% |
| 2013 | 5.61% | 2.38% | 2.307 | -1.17% | 64.02% |
| 2014 | 2.57% | 3.66% | 0.712 | -5.01% | 80.35% |
| 2015 | 3.91% | 2.46% | 1.575 | -1.06% | 64.59% |
| 2016 | 3.30% | 2.33% | 1.410 | -1.53% | 47.44% |
| 2017 | 1.64% | 2.05% | 0.804 | -1.35% | 61.08% |
| 2018 | 3.08% | 3.84% | 0.808 | -3.38% | 81.30% |
| 2019 | 12.30% | 3.64% | 3.206 | -1.03% | 80.89% |
| 2020 | 15.53% | 9.57% | 1.557 | -4.78% | 88.45% |
| 2021 | 10.83% | 5.96% | 1.755 | -5.47% | 96.18% |
| 2022 | 32.24% | 8.02% | 3.525 | -2.63% | 99.23% |
| 2023 | 9.70% | 5.32% | 1.768 | -4.85% | 96.07% |
| 2024 | -2.61% | 5.73% | -0.434 | -5.09% | 98.07% |

## D Appendix D: Output Manifest

| File | Purpose |
|---|---|
| outputs/performance_summary.csv | Headline 50M, 250M and 1B performance table |
| outputs/daily_returns_250m.csv | Daily 250M net returns used for QuantStats |
| outputs/positions_250m.parquet | Position-level economics, borrow tier, capacity |
| outputs/report_ready/feature_inventory.csv | Feature formulas, lags and observability |
| outputs/report_ready/locked_strategy_robustness.csv | Year, stress, tail and autocorrelation robustness |
| notebooks/c2o_research_process_and_tests.ipynb | Notebook that checks report numbers from outputs |

The repository contains more audit files, but these are directly tied to the report's headline claims. To reproduce, place original data files under `data/` and run `make reproduce`.